Assignment 2: KNN with CUDA

100 Points Possible

10/25/2026

Attempt 1

In Progress

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Unlimited Attempts Allowed

10/25/2026

Details

The assignment consists of implementing the KNN algorithm on the GPU using the CUDA programming model and comparing its performance with the CPU single-thread, multi-thread, and MPI implementations. Conduct all the code optimizations you consider relevant to speed up its execution, as long as the output is correct (accuracy of sequential / MPI / GPU must be the same). You can reuse the code from the previous assignment to extend it to the GPU.

Activities:

1. Implement a CUDA version of the KNN algorithm on the GPU.
2. Run the CUDA code on **maple.cs.vcu.edu** and **athena.hprc.vcu.edu** on the small, medium, and large datasets, respectively.
3. Analyze the performance using different kernel configurations (e.g. different number of threads per block, different code approaches to solving the problem with/without shared memory, etc.) to identify which approach is the most efficient. You can deliver different versions of the code with different strategies. Please make sure to describe in the report what are the differences among the versions.
4. Make the GPU code scalable to multiple GPUs using MPI.

Deliverables:

1. Source code. Provide the KNN implementation using CUDA. You can deliver several implementations, make sure to describe their differences.
2. Slurm scripts to run (correctly) the code on athena.hprc.vcu.edu.
3. Report with the methodology followed and results (including tables and figures).

Compress the source code (all files including code, scripts, datasets, and Makefile) and the Excel report into a .zip file. Upload the .zip file to Canvas.

Warning: cheating/plagiarism of the code will be prosecuted. You've been warned.

View Rubric

Assignment CUDA

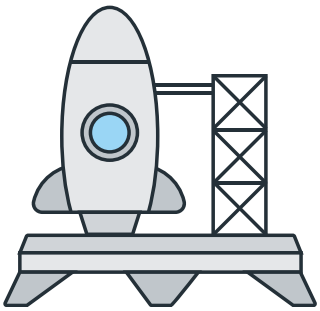
Criteria	Ratings			
Compiles and runs view longer description	**Full Credit** Both CUDA single-GPU and MPI+CUDA programs build using the submitted Makefile and run successfully on all three datasets **8 pts**	**Partial** Both programs build, but at least one fails to run successfully on one or more datasets **5 pts**	Exactly or successfu the other successfu	
Accuracy matches serial view longer description	**Full Accuracy** 6 of 6 combinations match. **14 pts**	**One Mismatch** 5 of 6 combinations match **12 pts**	**Two Mismatches** 4 of 6 combinations match **10 pts**	**Three Mismatches** 3 of 6 combinations match **7 pts**
CUDA implementation and optimizations view longer description	**Complete** Implements at least two distinct kernel strategies and describes their differences in the report. Explains meaningful optimizations, such as shared-memory reuse, global-memory access patterns, or stride-based indexing. **8 pts**	**Mostly Complete** Implements two kernel versions, but their differences or optimizations are minor or weakly justified. **6 pts**	Implemen kernel wit optimizati	
Maple results view longer description	**Verified** The implementation runs successfully on all three datasets, with accuracy and runtime output available for each **6 pts**		No usable	
Athena results (including Slurm scripts) view longer description	**Exceeds** All submitted CUDA versions complete three runs on all three datasets. Individual runtimes and averages are reported, and the Slurm scripts are present and correct. **10 pts**	**Mastery** Required runs and averages are available, but Slurm scripts are missing or contain errors. **7 pts**	**Near** Runs or averages are incomplete, or results for one dataset are missing. **4 p**	

Criteria	Ratings		
Kernel configuration experiments view longer description	**Complete** Threads-per-block is varied across values (e.g., 64/128). At least two code strategies are compared. Results are given for the small ,medium and large datasets, are averaged over 3 runs, and reported **12 pts**	**Mostly Complete** Both threads-per-block and code-strategy comparisons are present, but some runs are missing **9 pts**	Only one (threads-p is done ac
Analysis of kernel configurations view longer description	**Complete** Identifies the most efficient configuration and approach, supported by the data. Explains causes of the observed trends with GPU concepts **8 pts**	**Mostly Complete** Identifies the best approach and expla but causes are only partly or superficia	
Multi-GPU MPI+CUDA implementation view longer description	**Complete** Each MPI rank selects and uses its assigned GPU. Work is partitioned across ranks, and partial results are combined correctly. The program runs with one GPU and at least two multi-GPU configurations, producing accuracy that matches the serial CPU baseline. **8 pts**	**Mostly Complete** The implementation genuinely uses multiple GPUs but has a minor design or result-aggregation flaw. **6 pts**	MPI and C multiple G results are runs
Multi-GPU scaling experiments view longer description	**Complete** Tests at least three GPU counts on all datasets, with three runs and averages per configuration. Explains scaling trends and includes multi-node results or a documented limitation. **6 pts**	**Mostly Complete** Tests multiple GPU counts, but some runs, averages, or required multi-node evidence are missing. **4 pts**	Tests only reports re

Criteria	Ratings		
Speedup calculated correctly view longer description	**Full Credits** States the speedup formula, correctly calculates end-to-end speedup for all required results, compares GPU performance with serial CPU, Pthreads, OpenMP, and MPI implementations, and reports multi-GPU parallel efficiency. **4 pts**	**Partial** Speedup is calculated correctly, but CPU comparisons or multi-GPU efficiency are missing. **3 pts**	**S** Speedup errors or u
Methodology section view longer description	**Complete** Briefly describes the KNN algorithm. Explains the CUDA design, describes the differences among all delivered versions, explains the multi-GPU MPI partitioning, and states the hardware/software environment. **5 pts**	**Mostly Complete** Approaches are explained, but one required element is missing or unclear **3 pts**	Methodol required e
Results section (tables and figures) view longer description	**Complete** Presents all required results, including Athena runs, kernel configuration experiments, and multi-GPU scaling, using labeled tables and figures. **4 pts**	**Mostly Complete** All results are present, but some tables or figures have missing labels/units, or figures are missing **3 pts**	Multiple re
Report format view longer description	**Compliant** Report uses the required format throughout and follows the required structure (Introduction, Methodology, Results, Analysis, Conclusion) **2 pts**		Report do
All required files in zip view longer description	**Exceeds** All five categories are present with usable content. **5 pts**	**Mastery** One category is missing **4 pts**	**Near** Two categories are missing **3 p**

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